Operations Research for Health Care

Lesson 8/15

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UFMG, Belo Horizonte, Brasil
11.12.2024

Outline of this lecture

Efficiency vs Effectiveness

Access to health care

Need, Demand and Supply on health care

Measuring coverage, utilization, and quality on health care

Measuring spatial accessibility on health care

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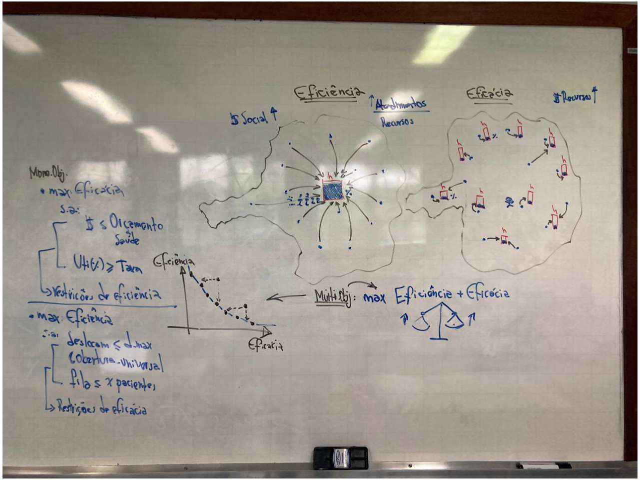
A framework for the study of Access to medical care [Simon, 1955]

- **Efficiency** (*doing things right*): achieving a goal with the least amount of resources (time, money, effort, etc.) (*economic rational man*).
- **Effectiveness** (*doing the right thing*): Achieving the desired goal or outcome, regardless of the resources used. (*behavourial man*).



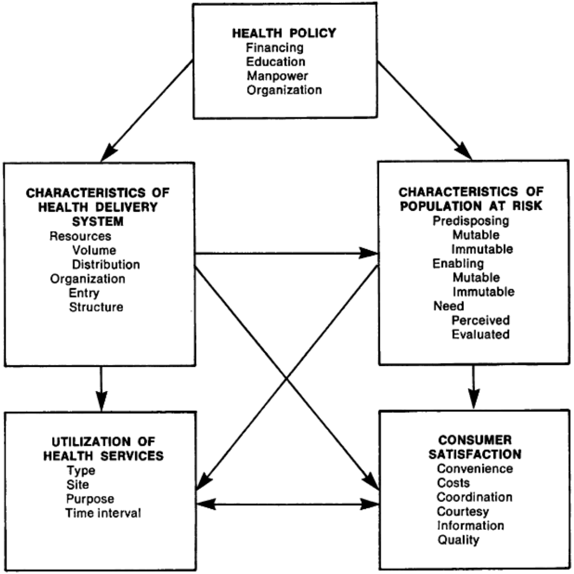
Outline of this lecture

A framework for the study of Access to medical care [Aday and Andersen, 1974]

Access is use, not just the presence of a facility. Access is measured by use relative to "need". Patients and professionals assess "needs" differently. There are two components to services use: "initiation" and "continuation". Barriers to access are financial, psychological, informational, social, organisational, spatial, temporal, and so on. - adapted from Avedis Donabedian in [Aday and Andersen, 1974].

- **All**: Policy makers, administrators, and patients: "access to health care system should be improved".
- **Goal**: Redistribution of resources.
- **Purpose**: Construct an integrated theoretical framework for health care access.
- **Defintions**: Asscessibility; a health policy framework; a delivery system; population at risk; and outcome indicators.

A framework for the study of Access to medical care [Aday and Andersen, 1974]



Framework for the study of access.

A framework for the study of Access to medical care [Aday and Andersen, 1974]

Motivation: Access has been more of a political than an operational idea. The concept of "access" is frequently discussed, however, much less how it might be measured and what methods should be used to evaluate it.

- Access: Implies entry to the health care system.
- Access: Weighted sum of the difference between the ideal and actual number of *services, personnel, and equipment* in a given community.
- Accessibility (socio-organizational): *resources* that facilitate the efforts of the patient to obtain care. E.g., the provider's *fee* scale and specialization.
- Accessibility (geographic): More than the mere availability of resources. It is a function of the **time** and **distance** that must be traversed to get care.
- Access: The patient "willingness" to seek care (*knowledge* about health care system + knowledge of his/her illness. Influence the utilization).
- Access: The quantity and quality of patients through the medical care system.
- Those persons that actually need medical care and receive it, or ease and convenience of getting a physician.

A framework for the study of Access to medical care [Aday and Andersen, 1974]

The framework: Health policy objectives:

1. **Inputs**: Characteristics of the health care system and of the populations at risk.
2. **Outputs**: Actual utilization of health care services and patients evaluation of it.

Delivery system: *The system is the unit of analysis, rather than the individual*.

- **Resources** (*volume and the distribution in an area*): The labor and capital devoted to health care: health personnel, structures, education, and the equipment in providing health services. **Measures**: number of physicians, of hospital beds, and of ambulances per unit of population and per unit of geographic area.
- **Organization** (*what the system does with its resources*): travel time, waiting time, entry, continues the treatment, referral. **Measures**: mean travel time, mean appointment waiting time, and mean office waiting time for service, mean response time from initial call for emergency service to ambulance arrival.

A framework for the study of Access to medical care [Aday and Andersen, 1974]

Population at risk: *(determinants of utilization): the individual rather than the system is the unit of analysis*.

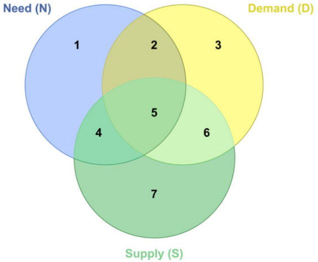
- Prior to the onset of illness episodes, i.e., *age, sex, race, religion, values concerning health and illness*.
- **Need component**: illness level, immediate cause of health service use, or what is evaluated by the delivery system.
- The household **survey** is the best method for collecting data on the population at risk. *Census, manpower data are best sources*.

Outcome Indicators

- Measuses by **type** of service used (e.g., hospital, physician, dentist, emergency care, home care), **the site** at which care was rendered (home, office, clinic, inpatient hospital, etc.), **the purpose** of the care received (preventive, curative, stabilizing, custodial), and **the time** interval involved (percent of population at risk who did and did not see a physician in a given time interval, mean number of visits to a physician in a given time interval).

Outline of this lecture

Recent definitions on Need, Demand and Supply [Santana et al., 2023]



1. There is Need – but neither Demand nor Supply.
2. There is Need and Demand – but not Supply.
3. There is Demand – but neither Need nor Supply
4. There is Need and Supply – but not Demand
5. There is Need, Demand and Supply
6. There is Demand and Supply – but not Need
7. There is Supply – but neither Need nor Demand

Recent definitions on Need, Demand and Supply [Santana et al., 2023]

- **[1]** Some people lack access to healthcare. This can be because they don't know they need it or because they know they need it but can't get it.
  - (**e.g.:** *(a) Unperceived unmet need: T is getting forgetful. T has dementia but been neither diagnosed nor treated.(b) Chosen unmet need: N has a condition she knows is treatable but she decides not to visit the GP about it.*)
- **[2]** Health care needs are expressed as demand, but supply does not meet demand. Therefore, there is no access to health care. In this case, health needs are unmet and 'supply constrained'. Possible causes include barriers to access, capacity constraints and waiting times.
  - (**e.g.:** *D has been depressed. Her low mood is not improving. She has asked the GP to refer her for psychotherapy. The GP has referred her but the waiting time is 6 months.*
- **[3]** Demand for health care is not linked to health care need and is not met by supply. There is no access to health care.
  - (**e.g.:** *Z has a painful hip. She asks her GP to refer her for an x-ray, but is refused because there is no evidence of arthritis. She also asks for a new medicine for her condition but is refused because the drug has not been proven to be cost-effective.*)

Recent definitions on Need, Demand and Supply [Santana et al., 2023]

- **[4]** Need meets supply and not demand is the natural place of prevention policies. There is access to health care.
  - (**e.g.:** *G's child's school says all pupils will get a vaccine next week.*)
- **[5]** Health care is effective: supply meets demand (based on need) and capacity to benefit is positive. There is access to health care.
  - (**e.g.:** *W has sinus problems. She visits her GP and is prescribed a nasal spray.*)
- **[6]** Supply meets demand but demand is not linked to health care need. There is utilisation but no access to health care.
  - (**e.g.:** *(a) Provider-induced demand: P's dentist calls her in for a check-up 6 months after her last one. She is in good dental health and only needs an annual check-up. (b) Patient-induced demand: N has a cold. Her GP diagnoses a respiratory virus, but prescribes antibiotics anyway.*)
- **[7]** There is excess capacity or inefficiencies in the delivery process. There is service availability but no access to health care. Supply is not utilised. Do not meed need or demand.
  - (**e.g.:** *In Hospital F, a third of outpatient appointments are "did not attend". Hospital F pays staff regardless.*)

Outline of this lecture

Efficiency vs Effectiveness

Access to health care

Need, Demand and Supply on health care

Measuring coverage, utilization, and quality on health care

Measuring spatial accessibility on health care

A conceptual framework for access, utilization, quality, and effective coverage [Shengelia et al., 2005]

A framework to clarify the inter-relationships between health care access, demand, utilization, and coverage.

Purpose: To explain factors that reveal *gaps in delivery system*. Highlight the *requirements* for health information systems.

- **Coverage**: The probability that individuals will receive health. Literature weakness: interventions are analyzed in isolation, but the availability of skilled human resources were not always recognized as system-wide constraints. E.g. educational planning for next 5-10 years.
- **Demand**: Factors that explain the volume of health care utilization and choice of provider (e.g., price, distance, perceived quality, insurance, etc). Modeled using economic models, e.g. *demand elasticity based on price, distance*.
- **Need**: health status, social vulnerability, and capacity to benefit from *interventions*.
- **Access**: *Supply* of services, but the quality of services delivered has rarely been incorporated into the analysis.

A conceptual framework for access, utilization, quality, and effective coverage [Shengelia et al., 2005]

Effective coverage: the probability that individuals will receive health gain from an intervention if they need it.

Integrating framework (6 issues):

1. **True need** (the potential to benefit from an intervention) can deviate from *perceived need*.
2. Intervention-specific **coverage**: for setting human resourses, financing, infrastructure.
3. **Quality**: capture aspects of making the *right diagnosis*.
4. Quantification of **demand** and **supply** side on the overall delivery system.
5. Develop a framework for **effective coverage** applicable at the individual and **population level**.
6. **Measure** the expectation of receiving potential health gain if needed.

A conceptual framework for access, utilization, quality, and effective coverage [Shengelia et al., 2005]

Effective coverage: quality, utilization, and need.

I : Individual.
J : Intervention
K : Provider

- EC_{ij}: Effective coverage for individual i with intervention j.
- Q_{ij}: (Quality) The expected quality of intervention j to be received by person i.
- U_{ij} (utilization) The probability that individual i will receive intervention j.
- N_{ij}: (need) The individual i's need for intervention j.

$$EC_{ij} = Q_{ij}U_{ij}|N_{ij} = 1 \tag{1}$$

A conceptual framework for access, utilization, quality, and effective coverage [Shengelia et al., 2005]

Effective coverage: quality, utilization, and need.

- HG_{ijk}: Health gain of i from intervention j from provider k.
- P_k: Provider k.
- P_k^{max}: Provider k with optimal perfomance.

$$Q_{ij} = \frac{\sum_{k=1}^{n} HG_{ijk}U_{ijk}}{\sum_{k=1}^{n} HG_{ijk}U_{ijk}|P_k = P_k^{max}} \tag{2}$$

EC_{ij} can be aggregated across individuals to yield an **overall EC** for a system.

$$EC_j = \frac{\sum_{i=1}^{n} EC_{ij}HG_{ij}Pr(N_{ij} = 1)}{\sum_{i=1}^{n} HG_{ij}Pr(N_{ij} = 1)} \tag{3}$$

$$EC = \frac{\sum_{j=1}^{n} EC_j HG_j|P_k = P_k^{max}}{\sum_{j=1}^{n} HG_j|P_k = P_k^{max}} \tag{4}$$

Effective Coverage - MathProgramming

```
set I;
set J;
set K;

param Ui{I,J,K};
param U{i in I, j in J}:= min{k in K}Ui[i,j,k];
param HGi{I,J,K};
param HGimax{i in I, j in J}:= max{k in K} HGi[i,j,k];
param HGij{i in I, j in J}:= sum{k in K}Ui[i,j,k]*HGi[i,j,k];
param HGmax{j in J}:= max{i in I}HGij[i,j];
param Q{i in I, j in J}:= HGij[i,j] / HGimax[i,j];
param N{I,J}, binary;

param ECi{i in I, j in J}:= Q[i,j]*U[i,j]*N[i,j];
```

Effective Coverage - MathProgramming

```
param ECjn{j in J}:= sum{i in I} ECi[i,j]*HGij[i,j]*N[i,j];
param ECjd{j in J}:= sum{i in I} HGij[i,j]*N[i,j];
param ECj{j in J}:= if ECjd[j] == 0 then 0 else ECjn[j] / ECjd[j];

param ECn:= sum{j in J} ECj[j]*HGmax[j];
param ECd:= sum{j in J} HGmax[j];
param EC:= ECn/ECd;


solve;
```

MathProgramming - Printing a Report

Good reports bring valuable information

```
printf "\n========= Utilization ========\n\t";
printf{j in J} "[%d]\t",j;
printf "\n";
for{i in I}{
printf "[%d]", i;
printf{j in J} "\t%.2f", U[i,j];
printf "\n";
}
printf "==============================\n\n";
printf "\n========= Health Gain ========\n\t";
printf{j in J} "[%d]\t",j;
printf "\n";
for{i in I}{
printf "[%d]", i;
printf{j in J} "\t%.2f", HGij[i,j];
printf "\n";
}
printf "==============================\n\n";
```

MathProgramming - Printing a Report

```
printf "\n=========== Quality ==========\n\t";
printf{j in J} "[%d]\t",j;
printf "\n";
for{i in I}{
printf "[%d]", i;
printf{j in J} "\t%.2f", Q[i,j];
printf "\n";
}
printf "==============================\n\n";
printf "\n===== Effective Coverage [i]==\n\t";
printf{j in J} "[%d]\t",j;
printf "\n";
for{i in I}{
printf "[%d]", i;
printf{j in J} "\t%.2f", ECi[i,j];
printf "\n";
}
printf "==============================\n\n";
```

MathProgramming - Printing a Report

```
printf "\n= POP Effective Coverage [j] =\n\t";
printf{j in J} "[%d]\t",j;
printf "\n";
printf " ";
printf{j in J} "\t%.2f", ECj[j];
printf "\n";
printf "==============================\n\n";
printf "=======================================\n";
printf "POPULATION EFFECTIVE COVERAGE: %.2f %%\n", EC*100;
printf "=======================================\n\n";
```

Outline of this lecture

Concepts and methods for PHC spatial accessibility [Guagliardo, 2004]

Primary care is the most effective form of healthcare for maintaining population health, as it is inexpensive, easily delivered, and highly effective at preventing disease.

1. Explicit definitions of access to care are not widely used to guide research.
2. The most measures of spatial accessibility to care lose validity in urban areas.

This paper:

- Explains basic concepts and measurements of **access**;
- Provides primary care researchers with some historical background (focus on urban areas: where is the major population and bad spatial analysis methods)
- Outlines the major questions of **geographic accessibility**

The *majority of research* and policy efforts **to improve access** and eliminate disparities in healthcare have focused on costs

Concepts and methods for PHC spatial accessibility [Guagliardo, 2004]

Access: (confusion) *Ability to get care? Act of seeking care? Actual delivery of care?*

- **Stages**:
  - Potential: *Needy population*
  - Realized: (actualized care, utilization) *Delivery of care*.
- **Dimensions** (5)
  - Aspatial: [1] *affordability*, [2] *acceptability* and [3] *accommodation* (health financing and cultural factors).
  - Spatial [**Spatial Accessibility (SA)**]: [4] *availability*, [5] *accessibility*
    - **Availability**: the number of local service points from which a client can choose.
    - **Accessibility**: travel impedance (distance or time) between patient location and service points.

Concepts and methods for PHC spatial accessibility [Guagliardo, 2004]

There is clear evidence of social inequalities in healthcare provision, including primary healh care (PHC), but little quantitative information on the impact of accessibility on population health. A long-standing question:

- – *What is the right rate of healthcare?*
- - *Is optimal PHC SA more important than the optimal specialists SA?*
- - *Is it more important than optimal SA of inpatient services?*

Measuring Spatial Accessibility (SA)

- **Supply ratios**: e.g. number of physicians, clinics, or hospital beds / population size within the area (census): *Minimal standards for unserved areas*.
- **Distance to nearest provider**: Or travel cost along road, rail, or a transportation network (choices limited: rural).
- **Average distance to providers** Measure of accessibility and availability.
- **Gravity models** Combined measure of accessibility and availability: measures take into account all alternative service points.

Concepts and methods for PHC spatial accessibility [Guagliardo, 2004]

The simplest formula for gravity-based accessibility, firts published by [Hansen, 1959]:

$$A_i = \sum_{j \in J} \frac{S_j}{d_{ij}^{\beta}} \qquad (1)$$

- A_i: Spatial accessibility from (origin) population point i (demand area).
- S_j: Service capacity at provider (destination) location j.
- d_{ij}: Travel impedance, e.g. distance or travel time, between points i and j.
- β: Distance decay, travel friction coefficient. *Require empirical investigation*.

Problems for health care:

1. A_i is *not intuitive to healthcare workforce policy makers*, who prefer to think of spatial accessibility in terms of *provider-to-population ratios or simple distance*.
2. *It only models supply*. There is no adjustment for demand. Then, [Joseph and Bantock, 1982] added demand adjustment V_j to de denominator.

Concepts and methods for PHC spatial accessibility [Guagliardo, 2004]

$$V_j = \sum_{k \in K} \frac{P_k}{d_{kj}^{\beta}} \tag{2}$$

- V_j: Demand on provider (destination) location j
- P_k: population size at (demand, origin) point k.
- d_{kj}: is the travel distance or travel time, between population (demand, origin) point k and (destination) j.

Improved gravity model for Accessibility of population i

$$A_i = \sum_{j \in J} \frac{S_j}{d_{ij}^{\beta} V_j} \tag{3}$$

- Recent developments: **Two-step floating catchment area**, **Compound gravity model**, **Kernel density method**.

Spatial Accessibility - MathProgramming

```
set I;
set J;
set K;

param Beta;
param Di{I,J}, >= 0;
param Dibeta{i in I, j in J}:= Di[i,j]^(Beta);
param S{J};
param A{i in I}:= sum{j in J}S[j]/Dibeta[i,j];

param P{K};
param Dk{K,J};
param Dkbeta{k in K, j in J}:= Dk[k,j]^(Beta);
param V{j in J}:= sum{k in K}P[k]/Dkbeta[k,j];

param Ai{i in I}:= sum{j in J}S[j]/(Dibeta[i,j]*V[j]);

solve;
```

MathProgramming - Printing a Report

```
printf "\n========= Accessibility Report ========\n";
printf "Beta:  %.2f\n", Beta;
printf "========================================\n";
printf "Dest\tSupply\tDemand\n";
for{j in J}{
printf "[%d]\t%4d\t%4d\n", j, S[j], V[j];
}
printf "========================================\n";
printf "Orig\tAccessibility\n";
for{i in I}{
printf "[%d]\t%.2f\n", i, Ai[i];
}
printf "========================================\n\n";
```

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